# Unitree Go2

## New Creature of Embodied AI

New and Improved
Returning with Glory

Infinite Revolution



# Standard Ultra-wide 4D LIDAR Upgrades Recognition System by 200%

Go2 features with Unitree's self-developed 4D LIDAR L1 with 360° x90° hemispherical ultra-wide recognition, super small blind spot and a minimum detection distance as low as 0.05m, which makes Go2 realize all-terrain recognizing.




**360°×90°**
Ultra-wide Scanning

**0.05m**
Blind Spot
(Radar accuracy decreases at close range detection)

**20m**
@90% Reflectivity

**21600 points/s**
Effective Frequency

**43200 points/s**
Frequency of Sample

**100Klux**
Anti-highlight Protection

※Laser Safety Class: Class 1(IEC60825-1:2014) Laser Safety

# Your New Intelligent Friend

### Tracking Module
Remote-controlled or
automatic tracking

### Front Camera
Image tansmission Resolution 1280*720
FOV 120 °
Ultra wide angle lens deliver rich clarity

### Front Lamp
Brightly lights the
way ahead

### Intercom Microphone
Effective communication with
no scenario restrictions

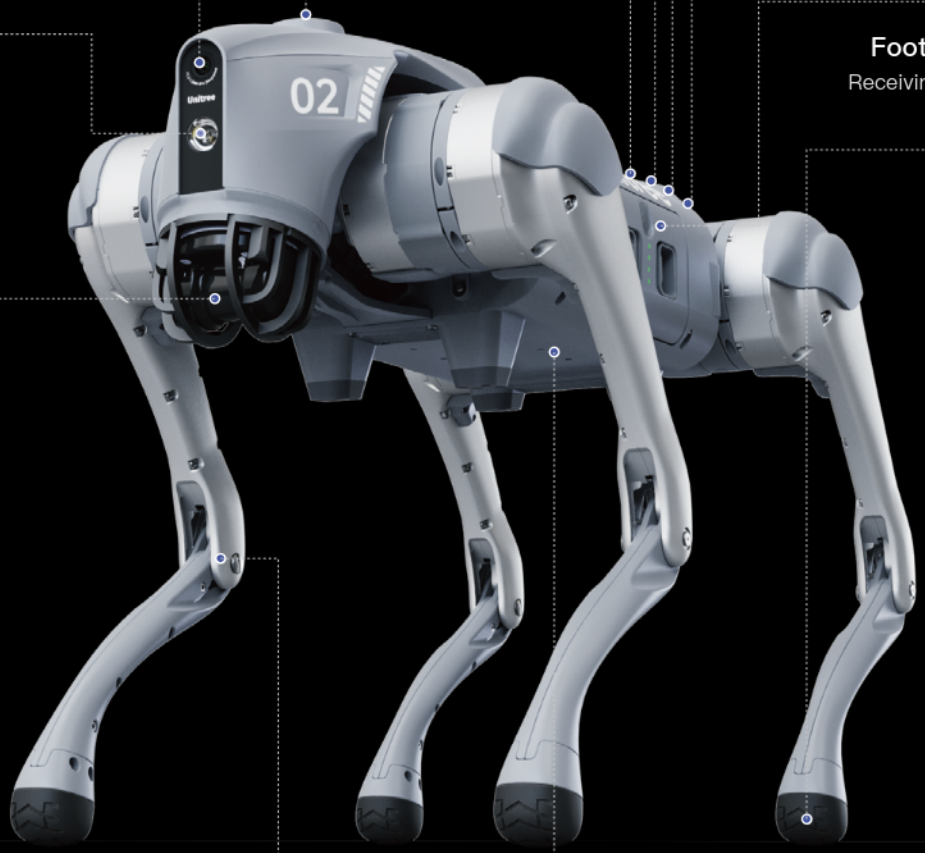
### Self-retracting Strap
Easy to carry and
load things

### Powerful Core
·Motion controller
High-performance ARM processor
·Improved AI algorithm processor
External ORIN NX/NANO

### Hardware Upgrade
·3D LIDAR
·4G ESIM Card
·WiFi6 with Dual-band
·Bluetooth 5.2
for stable connection
and remote control

### Smart Battery
Standard 8000m Ah battery
Long-endurance 15000m Ah battery
Protection from over-temp,
overcharge and short-circuit

### Foot Force Sensor
Receiving foot perception
in real time




### 4D LiDAR L1
360°*90° omnidirectional
ultra-wide-angle scanning allows
automatic avoidance with small blind
spot and stable operation

### 12 Knee Joint Motors
Strong and powerful
Beautiful and simple
Brandy new visual experience

### Speaker for Misic Play
listen to music as your pleasure

# New Intelligence——Unitree Go2



## Intelligent Side-follow System 2.0

By adopting the new wireless vector positioning and control technology, the positioning accuracy is technically upgraded by 50%, the remote control distance is over 30m[1], and combined with the optimised obstacle avoidance strategy, it can make the robot better traverse complex terrain.

[1] In open spaces with no shelter



## Motor performance enhanced by 30%

Go2 boasts a peak joint torque of 45N.m[2], a new internal trace connecting technique, and heat pipe coolers to decrease temperature effectively.

[2] The maximum torque in the table refers to the maximum torque of the largest joint motor; the actual maximum torque varies for the 12 joint motors.



## Battery capacity and endurance upgraded by 150%

Go2 is equipped with a battery capacity increased to 8,000mAh, as a 15,000mAh ultra-long life battery is optional, and a voltage increased to 28.8V to improve motor efficiency, power and stability.

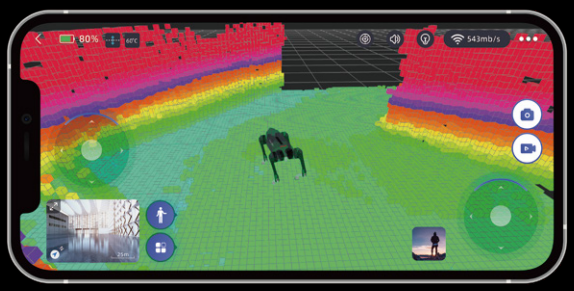


## Various actions and poses

Go2 boasts a variety of poses such as jumping, stretching, shaking hands, cheering, pouncing, and sitting down.

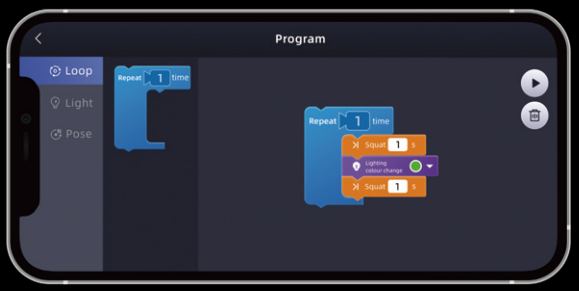
# Intelligent Interaction

Have great fun with the APP



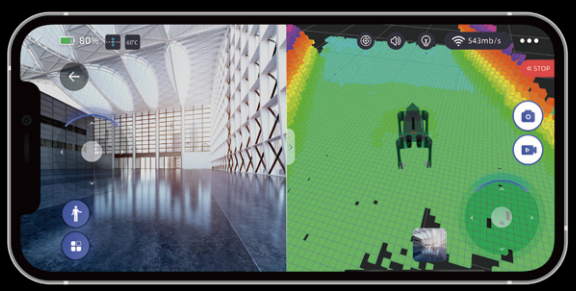
## Intelligent Avoidance
Precise and Agile

Equipped with 4D LiDAR L1, the robodog de-tects, captures and draws the 3D real world for user.




## Graphical Programming
Simple Yet Smarter

Optimise the graphical programming function, make it easy to complete the program design by simple drag, drop and connection.Make pro-gramming beginners easy to start and innovate.



## Hd Picture Quality
Real-time and Stable [1]

A new App realizes HD image transmission and real-time remote monitor.Built-in 4G and eSIM enables more stable connection and remote control.

[1]Transformation and quality varies considerably in different network environments.




## OTA Upgrades
Keep Improving and Evolving to be Smarter

With user authorisation, the robot automatically connects to a cloud-based OTA service to up-grade its own programs to continuously im-prove the user experience.



# Parameters

| | Type | AIR | PRO | EDU |
|---|---|---|---|---|
| Mechanical & Electron | Stand Height | 70×31×40cm | | |
| | Weight (With Battery) | About 15kg | | |
| | Material | Aluminium alloy + High strength engineering plastic | | |
| | Voltage | 28V~33.6V | | |
| | Peaking Capacity | About 3000W | | |
| Performance | Payload | ≈7kg（MAX ~ 10kg） | ≈8kg（MAX ~ 10kg） | ≈8kg（MAX ~ 12kg） |
| | Speed | 0~2.5m/s | 0~3.5m/s | 0~3.7m/s（MAX~5m/s） |
| | Max Climb Drop Height | About 15cm | About 16cm | |
| | Max Climb Angle | 30° | 40° | |
| | Basic Computing Power | ○ | 8-core High-performance CPU | |
| Joint | Peak Joint Torque [1] | ○ | About 45N.m | |
| | Range of motion | Body：-48~48° | Thigh：-200°~90° | Shank：-156°~-48° |
| | Intra-joint circuit (knee) | ● | ● | ● |
| | Joint Heat Pipe Cooler | ● | ● | ● |
| Force Sensor | Super-wide-angle 3D LIDAR | ● | ● | ● |
| | Wireless Vector Positioning Tracking Module | ○ | ● | ● |
| | HD Wide-angle Camera | ● | ● | ● |
| | Foot-end force sensor | ○ | ○ | ● |
| Feature List | Basic Action | ● | ● | ● |
| | Auto-scaling strap | ○ | ● | ○ |
| | OTA upgrades | ● | ● | ● |
| | RTT2.0 Image transmission | ● | ● | ● |
| | Graphical programme | ● | ● | ● |
| | Front lamp | ● | ● | ● |
| | WiFi6 with Dual-band | ● | ● | ● |
| | Bluetooth 5.2/4.2/2.1 | ● | ● | ● |
| | 4G | ○ | ● | ● |
| | Voice Function [2] | ○ | ● | ● |
| | ISS 2.0 | ○ | ● | ● |
| | Intelligent detection and avoidance | ● | ● | ● |
| | Charging Pile Compatibility | ○ | ○ | ● |
| | Secondary development [3] | ○ | ○ | ● |
| Accessories | Manual controller | Optional | | Standard |
| | High computing power modular | ○ | | Optional Nvidia Jetson Orin |
| | Smart battery | Standard（8000mAh） | | Long endurance（15000mAh） |
| | Endurance | About 1-2h | | About 2-4h |
| | Charger | Standard（33.6V 3.5A） | | Fast charge（33.6V 9A） |

※The above parameters may vary in different scenarios and configurations, please subject to actual situations.
If any change in the appearance of the product, please refer to the actual product.
[1] The maximum torque in the table refers to the maximum torque of the largest joint motor; the actual maximum torque varies for the 12 joint motors.
[2] Voice functions include offline voice interaction, commands, intercom and music play.
[3] For more information, please read the secondary development manual.

# Extensions

## LIDAR



| | |
|---|---|
| Model | XT16 |
| Size (Without Bracket) | Φ100.0 / 103.0 mm*76mm |
| Voltage Range | 9–36V DC |
| Laser Wavelength | 905mm |
| Fov | Horizontal 360°, Vertical 30° (–15°~+15°) |

## LIDAR



| | |
|---|---|
| Model | MID–360 |
| Size (without bracket) | 65mm*65mm*60mm |
| Voltage range | 9–27V DC |
| Laser wavelength | 905mm |
| FOV | Horizontal 360°, Vertical–7°~52° |

## Depth Camera



| | |
|---|---|
| Model | D435i |
| Size | 124mm*29mm*26mm |
| Min Depth Distance | 0.105m |
| Depth Image Resolution | 1280*720 @ 30fps;<br>848*480 @ 90 fps |
| Depth Field Of View | 86° * 57° (±3°) |

## High Computing Power Module



| | |
|---|---|
| Model | Orin Nano 8GB、Orin NX 16GB |
| Voltage range | 16–60V DC |
| Computing power | Nano supports up to 40 Tops<br>NX supports up to 100 Tops |
| Expansion interface | USB3.0–Type A X1<br>USB3.0–Type C X2<br>Gigabit Ethernet port (standard RJ45) X2<br>100Gb Ethernet (GH1.25–4PIN) X1<br>M8 Air Plug Interface X1 |

## Compact Servo Arm



| | |
|---|---|
| Model | D1 |
| Degree Of Freedom | 6 |
| Playload | About 500g |
| Max Armspan | 550mm (without jaws) |
| Repeated Positioning Accuracy | About 0.2mm |
| Power Requirement | 24V 2.5A (MAX 5A) |
| Interface | DC5.5–2.1 |
| Motor Type | Servo |
| Power | 60W |
| Control Interface | Control communication interface RJ45 (ETH) |

## Remote Control with Screen 3 In 1



| | |
|---|---|
| Number of cameras | 2 |
| Camera Resolution | 1920x1080 |
| Wireless frequency | 2.4GHz |
| Searchlight Power | 30W |
| Horn power | 30W |
| Alarm light | Red and blue sharp–flash |
| Remote control with screen | MK15 |




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